

Why We Age

Is ageing inevitable — or is it simply a biological programme that evolution has chosen to run?

wear and tear · thermodynamics · dynamic equilibrium · mutations · metabolic rate · immortality · genetically determined · biological clock · hibernation · lethargy · cardiovascular · inexorable · unicellular organisms · life span · frugally · evolution · adaptation · captivity · equable

PART ONE · OBJECTS AGE DIFFERENTLY FROM ORGANISMS

- A** Our **life span** is restricted. Everyone accepts this as 'biologically' obvious. 'Nothing lives for ever!' However, in this statement we think of artificially produced, technical objects, products which are subjected to natural **wear and tear** during use. This leads to the result that at some time or other the object stops working and is unusable — 'death' in the biological sense. But are the wear and tear and loss of function of technical objects and the death of living organisms really similar or comparable?
- B** Our 'dead' products are 'static', closed systems. It is always the basic material which constitutes the object and which, in the natural course of things, is worn down and becomes 'older'. Ageing in this case must occur according to the laws of physical chemistry and of **thermodynamics**. Although the same law holds for a living organism, the result of this law is not **inexorable** in the same way. At least as long as a biological system has the ability to renew itself it could actually become older without ageing; an organism is an open, dynamic system through which new material continuously flows. Destruction of old material and formation of new material are thus in permanent **dynamic equilibrium**. The material of which the organism is formed changes continuously. Thus our bodies continuously exchange old substance for new, just like a spring which more or less maintains its form and movement, but in which the water molecules are always different.

"An organism is an open, dynamic system through which new material continuously flows. Destruction of old material and formation of new material are thus in permanent dynamic equilibrium — like a spring whose water molecules are always different, yet whose form and movement remain the same."

■ TECHNICAL OBJECTS — HOW THEY AGE

- Static, closed systems — the same material throughout their life.
- Subject to **wear and tear** according to the laws of physical chemistry.
- Ageing is **inexorable** — governed by **thermodynamics**, it cannot be reversed.
- No ability to repair or renew themselves — damage simply accumulates.
- 'Death' occurs when the material degrades beyond the point of function.

■ LIVING ORGANISMS — HOW THEY AGE

- Open, dynamic systems — new material continuously flows through them.
- Possess many mechanisms for self-repair at cellular and molecular level.
- Ageing is not **inexorable** — the organism can renew itself while remaining alive.
- Old material is destroyed and new material formed in **dynamic equilibrium**.
- In principle, a biological system need not age — even as it becomes older.

PART TWO · WHY DEATH IS NECESSARY FOR LIFE

C Thus ageing and death should not be seen as inevitable, particularly as the organism possesses many mechanisms for repair. It is not, in principle, necessary for a biological system to age and die. Nevertheless, a restricted **life span**, ageing, and then death are basic characteristics of life. The reason for this is easy to recognise: in nature, the existent organisms either adapt or are regularly replaced by new types. Because of changes in the genetic material — **mutations** — these have new characteristics and in the course of their individual lives they are tested for optimal or better **adaptation** to the environmental conditions. **Immortality** would disturb this system — it needs room for new and better life. This is the basic problem of **evolution**.

◆ Key Insight — Why Immortality Would Be Catastrophic for Evolution

If organisms did not die, there would be no room for new organisms with better-adapted **mutations** to take their place. **Evolution** depends on the constant testing and replacing of organisms over generations. An immortal organism would occupy resources and space indefinitely, preventing the emergence of any new or better-adapted life. Death is not a flaw in the system — it is the engine that drives evolutionary progress forward.

PART THREE · LIFE SPAN AS A BIOLOGICAL CHARACTERISTIC

D Every organism has a **life span** which is highly characteristic. There are striking differences in life span between different species, but within one species the parameter is relatively constant. For example, the average duration of human life has hardly changed in thousands of years. Although more and more people attain an advanced age as a result of developments in medical care and better nutrition, the characteristic upper limit for most remains 80 years. A further argument against the simple **wear and tear** theory is the observation that the time within which organisms age lies between a few days — even a few hours for **unicellular organisms** — and several thousand years, as with mammoth trees.

E If a **life span** is a **genetically determined** biological characteristic, it is logically necessary to propose the existence of an internal clock, which in some way measures and controls the ageing process and which finally determines death as the last step in a fixed programme. Like the life span, the **metabolic rate** has for different organisms a fixed mathematical relationship to the body mass. In comparison to the life span this relationship is 'inverted': the larger the organism the lower its metabolic rate. Again this relationship is valid not only for birds, but also, similarly on average within the systematic unit, for all other organisms — plants, animals, and **unicellular organisms**.

80 yrs

characteristic upper
limit of human life span

1000s yrs

life span of mammoth
trees — longest-lived
organisms

10%

longer women live
compared to men on
average

2×

longer mice live on
low-calorie diets

PART FOUR · ENERGY, METABOLISM AND LONGEVITY

F Animals which behave '**frugally**' with energy become particularly old — for example, crocodiles and tortoises. Parrots and birds of prey are often held chained up. Thus they are not able to 'experience life' and so they attain a high **life span** in **captivity**. Animals which save energy by **hibernation** or **lethargy** — such as bats or hedgehogs — live much longer than those which are always active. The **metabolic rate** of mice can be reduced by a very low consumption of food, a hunger diet. They then may live twice as long as their well-fed comrades. Women become distinctly — about 10 per cent — older than men. If you examine the metabolic rates of the two sexes you establish that the higher male metabolic rate roughly accounts for the lower male life span. That means that men live life 'energetically' — more intensively, but not for as long.

G It follows from the above that sparing use of energy reserves should tend to extend life. Extreme high-performance sports may lead to optimal **cardiovascular** performance, but they quite certainly do not prolong life. Relaxation lowers **metabolic rate**, as does adequate sleep and in general an **equable** and balanced personality. Each of us can develop his or her own 'energy saving programme' with a little self-observation, critical self-control and, above all, logical consistency. Experience will show that to live in this way not only increases the **life span** but is also very healthy. This final aspect should not be forgotten.

◆ Practical Conclusion — Your Personal Energy-Saving Programme

The evidence from biology is consistent: organisms that use energy more sparingly tend to live longer. For humans, this translates into a clear set of practical recommendations: get adequate sleep, cultivate an **equable** and balanced personality, practise regular relaxation, and avoid the kind of extreme physical exertion that burns energy at maximum rate without adding years to your life. Paradoxically, the path to a longer life may be less about working harder — and more about working less.

■ TALKING POINTS

The article argues that death is not a flaw but necessary for evolution. Do you find this argument convincing? Why?

The article suggests that extreme sport does not prolong life. Does this surprise you? How do you reconcile this with common health advice?

If immortality were scientifically possible, would it be a good thing for humanity? What problems might it cause?

The article says an equable and balanced personality helps people live longer. Do you think personality can really be changed — or is it fixed?